Fine-Grained Dog Breed Classification on Stanford Dogs: A Reproducible Transfer-Learning Baseline and Error Analysis

Author: [Add author name and affiliation]

Literature Review

The Stanford Dogs benchmark was designed to expose the effects of low inter-class differences and high within-class variation, including pose, occlusion, and background clutter [1]. Early fine-grained work explored discriminative local patches and randomized decision trees [3]. Sermanet et al. applied recurrent visual attention directly to Stanford Dogs without bounding-box supervision. Their three-resolution, one-glimpse model reported 76.3% mean accuracy, compared with 75.5% for a GoogLeNet baseline [4]. These are mean-accuracy results and should not be treated as identical to top-1 accuracy without checking the evaluation definitions.

Bilinear CNNs capture local pairwise feature interactions by combining two convolutional feature streams [5]. Fully Convolutional Attention Networks (FCAN) learn discriminative image regions and report 87.3% for a fine-tuned ResNet-50 baseline, 88.5% with one attention region, and 88.9% with two on Stanford Dogs [7]. NTS-Net introduces a Navigator, Teacher, and Scrutinizer framework for locating informative regions on other fine-grained benchmarks; its reported experiments do not include Stanford Dogs [8].

Vision Transformers represent images as sequences of patches and can transfer effectively from large-scale pretraining [9]. Conde and Turgutlu use attention-guided cropping and erasing and report 93.2% accuracy on Stanford Dogs [10]. TransFG uses attention to select discriminative patches and a contrastive objective; it reports 92.3% on Stanford Dogs, compared with 91.7% for its standard ViT baseline [11]. Cui et al. study a separate pipeline that combines features from four CNNs, feature selection, and an SVM [12]. These results motivate fine-grained and attention-based extensions, but direct numerical comparison requires matching the split, image resolution, pretraining, and evaluation protocol.

Gap Analysis

The literature already includes attention-based and transformer-based methods with substantially higher reported accuracy than a small ResNet baseline. This project does not claim that ResNet-18 is a new method or that the chosen analysis is absent from prior work. Its project-level contribution is a documented baseline using the official split, macro-averaged metrics, per-breed results, and an error analysis that can be reproduced from the supplied archive.

The experiment addresses two questions:

1. How well does an ImageNet-pretrained ResNet-18 classify all 120 breeds under the official Stanford Dogs split?
2. Which breeds and breed pairs account for the largest share of its classification errors?

Findings

The selected ResNet-18 checkpoint achieved 66.08% top-1 test accuracy and 64.92% macro F1 on 8,580 official test images. Best validation accuracy was 65.50% at epoch five. The weakest recall was miniature poodle (12.7%). The results show overfitting after the validation peak and recurring confusion among visually similar breeds.

Methodology

4.1 Dataset and split

The local `archive.zip` contains 20,580 images and 20,580 annotation files. Stanford's official `train_list.mat` and `test_list.mat` were obtained separately from the dataset website. The official split contains 12,000 training images and 8,580 test images [1]. A stratified 10% subset of the official training partition (1,200 images) was used for validation, leaving 10,800 images for model fitting. The official test partition was not used for training or checkpoint selection.

4.2 Preprocessing and model

Images were processed at 224 × 224 pixels. Training augmentation used random resized crops (scale 0.7–1.0), horizontal flips, and mild brightness, contrast, and saturation jitter. Validation and test images were resized to 256 pixels and center-cropped to 224 pixels. All images were normalized with ImageNet channel means and standard deviations.

The classifier used ImageNet-pretrained ResNet-18 weights. Its final fully connected layer was replaced by a 120-output layer. All model parameters were fine-tuned using cross-entropy loss and AdamW with a learning rate of 3×10^{-4} and weight decay of 10^{-4}. Training used batches of 32 for ten epochs with random seed 42. The run used Apple's MPS backend. The checkpoint with the highest validation accuracy was retained and evaluated once on the official test set.

4.3 Evaluation

The test evaluation reports top-1 accuracy, macro precision, macro recall, macro F1, a 120 × 120 confusion matrix, and per-breed precision, recall, F1, and support. Macro averaging gives each breed equal weight. The plotted confusion subset highlights the 12 breeds with the largest number of errors. The experiment is a single run; no confidence interval or multi-seed average is reported.

Further Discussion

The overall test accuracy is substantially below results reported for larger fine-grained systems such as TransFG and the attention-guided ViT approach of Conde and Turgutlu [10], [11]. These are contextual reference points rather than direct head-to-head comparisons: this project uses a smaller ResNet-18, a 224-pixel input, a different training recipe, and only one run. The key finding from this experiment is the gap between 93.3% training accuracy and the 65.5% best validation accuracy, together with the validation decline after epoch five. This suggests that stronger regularization, a learning-rate schedule, early stopping, or more suitable fine-grained features should be investigated.

The per-breed results show why macro metrics and confusion analysis matter. Overall accuracy alone would hide the poor recall for miniature poodle and Rhodesian ridgeback.

Several prominent errors involve breeds with similar appearance or related shape, but visual inspection is needed before attributing the errors to specific features. Bounding-box annotations could support a controlled follow-up study comparing full-image classification with dog-centered crops.

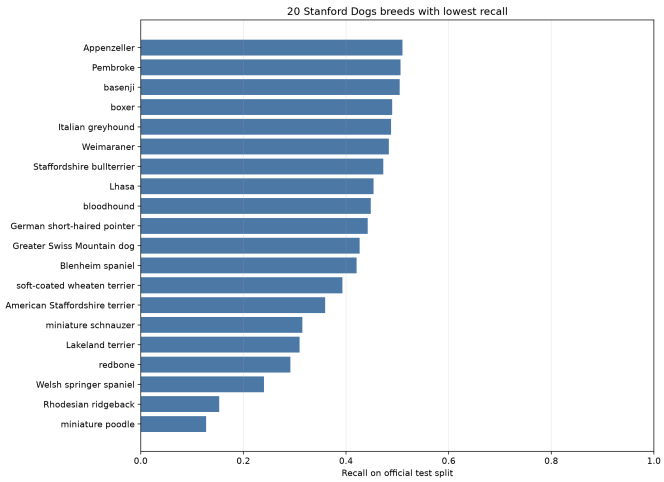
Important limitations are the single random seed, the single model family, and the absence of an ablation study. The project also does not measure inference latency or compare accuracy under a fixed compute budget. Consequently, it supports a baseline and error analysis, not a claim of state-of-the-art performance or broad generalization to photographs outside Stanford Dogs.

Results

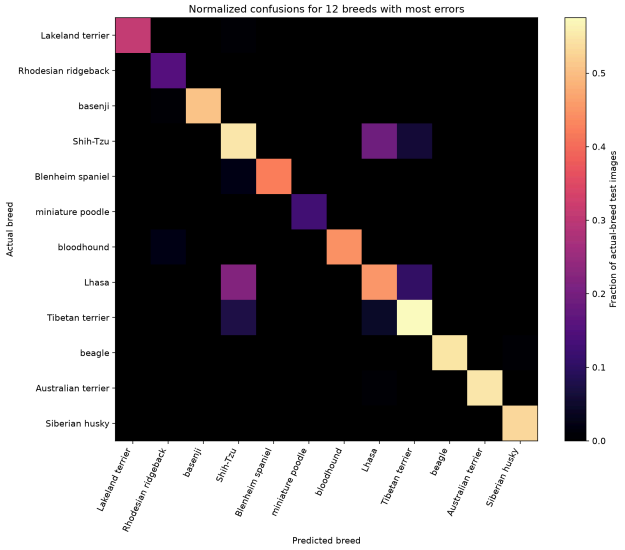
The best validation accuracy was 65.50% at epoch five. The selected checkpoint achieved 66.08% top-1 test accuracy, 68.24% macro precision, 65.57% macro recall, and 64.92% macro F1 on the 8,580 official test images. At epoch ten, training accuracy had reached 93.3% while validation accuracy was 59.7%, consistent with overfitting after the validation peak.

The weakest test recall was for miniature poodle (12.7%), followed by Rhodesian ridgeback (15.3%) and Welsh springer spaniel (24.0%). Frequent directed confusions included Blenheim spaniel → Japanese spaniel (36 images), miniature poodle → toy poodle (29), silky terrier → Yorkshire terrier (25), and miniature schnauzer → standard schnauzer (24). The errors suggest that the model struggles with several visually related breed pairs, although a confusion count alone does not explain the visual cause.

Figure 1 shows the 20 breeds with lowest test recall. Figure 2 shows row-normalized confusions for the 12 breeds with the most errors.



Twenty breeds with lowest recall



Normalized confusion matrix for high-error breeds

Conclusion

An ImageNet-pretrained ResNet-18 was trained on the Stanford Dogs official training partition and evaluated on its untouched official test partition. It achieved 66.08% top-1 accuracy and 64.92% macro F1. The test results revealed substantial class-level variation and recurring confusions among similar breeds. The training run also showed overfitting after its fifth-epoch validation peak. Future work should compare a stronger fine-grained model under the same split, run multiple seeds, and test whether bounding-box crops or regularization improve the weakest classes.

References

[1] A. Khosla, N. Jayadevaprakash, B. Yao, and L. Fei-Fei, "Novel Dataset for Fine-Grained Image Categorization: Stanford Dogs," in *First Workshop on Fine-Grained Visual Categorization, IEEE CVPR*, 2011. https://people.csail.mit.edu/khosla/papers/fgvc2011.pdf

[2] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A Large-Scale Hierarchical Image Database," in *IEEE CVPR*, 2009. https://image-net.org/static_files/papers/imagenet_cvpr09.pdf

[3] B. Yao, A. Khosla, and L. Fei-Fei, "Combining Randomization and Discrimination for Fine-Grained Image Categorization," in *IEEE CVPR*, 2011. https://svl.stanford.edu/assets/publications/pdfs/YaoKhoslaFeiFei_CVPR2011.pdf

[4] P. Sermanet, A. Frome, and E. Real, "Attention for Fine-Grained Categorization," *ICLR Workshop*, 2015. https://arxiv.org/abs/1412.7054

[5] T.-Y. Lin, A. RoyChowdhury, and S. Maji, "Bilinear CNN Models for Fine-Grained Visual Recognition," in *IEEE ICCV*, pp. 1449–1457, 2015. https://openaccess.thecvf.com/content_iccv_2015/html/Lin_Bilinear_CNN_Models_ICCV_2015_paper.html

[6] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in *IEEE CVPR*, pp. 770–778, 2016. https://openaccess.thecvf.com/content_cvpr_2016/html/He_Deep_Residual_Learning_CVPR_2016_paper.html

[7] X. Liu, T. Xia, J. Wang, Y. Yang, F. Zhou, and Y. Lin, "Fully Convolutional Attention Networks for Fine-Grained Recognition," in *IEEE CVPR*, 2016. https://arxiv.org/abs/1603.06765

[8] Z. Yang, T. Luo, D. Wang, Z. Hu, J. Gao, and L. Wang, "Learning to Navigate for Fine-Grained Classification," in *European Conference on Computer Vision (ECCV)*, pp. 420–435, 2018. https://openaccess.thecvf.com/content_ECCV_2018/html/Ze_Yang_Learning_to_Navigate_

[9] A. Dosovitskiy et al., "An Image Is Worth 16×16 Words: Transformers for Image Recognition at Scale," in *International Conference on Learning Representations (ICLR)*, 2021. https://openreview.net/forum?id=YicbFdNTTy

[10] M. V. Conde and K. Turgutlu, "Exploring Vision Transformers for Fine-Grained Classification," in *IEEE CVPR Workshops (FGVC8)*, 2021. https://arxiv.org/abs/2106.10587

[11] J. He, J.-N. Chen, S. Liu, A. Kortylewski, C. Yang, Y. Bai, and C. Wang, "TransFG: A Transformer Architecture for Fine-Grained Recognition," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 1, pp. 852–860, 2022. https://ojs.aaai.org/index.php/AAAI/article/view/19967

[12] Y. Cui, B. Tang, G. Wu, L. Li, X. Zhang, Z. Du, and W. Zhao, "Classification of Dog Breeds Using Convolutional Neural Network Models and Support Vector Machine," *Bioengineering*, vol. 11, no. 11, 1157, 2024. https://doi.org/10.3390/bioengineering11111157

Abstract

This study evaluates an ImageNet-pretrained ResNet-18 for 120-class dog-breed classification on the Stanford Dogs dataset. The model was trained with a fixed official split, using 10% of the official training partition for validation and reserving the official test partition for final evaluation. The best validation checkpoint was selected from ten epochs of fine-tuning. On the 8,580-image test partition, the model achieved 66.08% top-1 accuracy, 68.24% macro precision, 65.57% macro recall, and 64.92% macro F1. Errors were concentrated in visually similar breeds: miniature poodles were recalled at 12.7%, and frequent confusions included Blenheim spaniels with Japanese spaniels and miniature poodles with toy poodles. Training accuracy continued to increase after validation performance peaked at epoch five, indicating overfitting under this configuration. These results provide a reproducible baseline and class-level error analysis; they do not establish a new state of the art or a novel architecture.

Keywords: fine-grained visual classification, dog-breed recognition, transfer learning, ResNet-18, Stanford Dogs, error analysis

Introduction

Fine-grained image classification distinguishes categories with small visual differences. Dog breeds are a clear example: two breeds may share coat color or body shape, while one breed can vary substantially in age, pose, grooming, and appearance. The Stanford Dogs dataset was introduced for this problem and contains 120 breeds, 20,580 images, class labels, and bounding-box annotations [1].

A classifier that predicts the correct breed across a full 120-class benchmark is a useful computer-vision baseline. However, one overall accuracy number can conceal severe differences among breeds. A model can perform well on many classes while confusing visually similar pairs or failing on a small subset. This project therefore evaluates both overall performance and per-breed errors.

The experiment uses an ImageNet-pretrained ResNet-18, with its final layer replaced for the 120 dog breeds. ResNet provides a widely used transfer-learning baseline [6]. The goal is to document a reproducible training procedure, report results on Stanford's official test split, and identify class confusions that can guide later experiments.

Keywords: fine-grained visual classification, dog-breed recognition, transfer learning, ResNet-18, Stanford Dogs, error analysis